

Assessing Individual Decision-Making Skill by Manipulating Predictive and Unpredictive Cues in a Virtual Baseball Batting Environment

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ABSTRACT

We propose a virtual reality (VR) baseball batting system for assessing individual decision-making skill based on swing judgement of pitch types and the underlying prediction ability by manipulating combinations of pitching motion and ball trajectory cues. Our analysis of data from 10 elite baseball players revealed highly accurate swing motions in conditions during which the batter made precise swing decisions. Delays in swing motion were observed in conditions during which predictive cues were mismatched. Our findings indicated that decision-making based on pitch type influences the inherent stability of decision and accuracy of hitting, and that most batters made decisions based on pitching motion cues rather than on ball trajectory.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Human behavior depends on the integration of decision-making and motor control processes. In the context of baseball, batting provides a good example of the mutual interaction between decision-making and motor control under strict time constraints; players must make appropriate decisions regarding whether to hit the ball and swing precisely immediately thereafter. Despite the high degree of uncertainty and severe time constraints of the hitting situation, batters can use pitching motion cues to predict pitch type and trajectory to make quick yet accurate decisions. Indeed, researchers have reported that the ability to identify pitch type and pitch location at an early stage improves batting performance in terms of, scoring rate and walk/strikeout ratio in official games [1]. Catchers have also been reported to predict the trajectory of the pitch based on the pitcher's motion [4]. Kobayashi and Kimura [3] analyzed the interaction between decision-making and motor processes under time constraints using a task that imitates hitting in baseball using reaching movements, suggesting that decision-making influences sports performance. However, these studies were conducted by presenting pitching images in a 2D display, and it has been difficult to clarify the effect of information related to pitching motion and decision-making on actual batting. To address this issue, we designed a virtual reality (VR) environment in which intensive, repeatable manipulation of visual conditions can be performed in a realistic setting. Recent studies have also used VR systems for sports

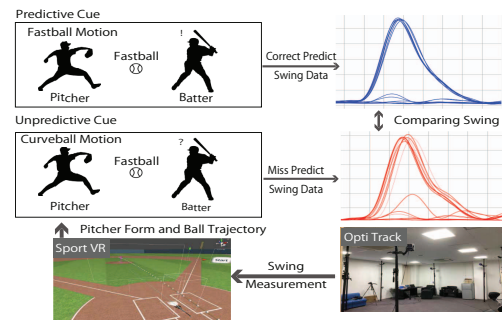


Figure 1: Overview of experimental conditions

training and skills assessment [2]. In this study, we used our VR batting baseball system to evaluate the decision-making skills of batters and underlying predictive ability based on pitching motion cues using a motion capture system (see Figure 1).

2 EXPERIMENT

Our baseball VR system allows batting in a virtual baseball field implemented in Unity. For pitching stimuli, we used pitching data (pitching motion and ball trajectory) from former professional baseball players, recorded using motion capture and the Trackman system. The pitcher's motion was modeled using biological motion to display information related to pitching motion only. The system can present a pitch with the trajectory of a fastball and motion cues associated with a curveball. If the batter predicts a curveball based on the pitching motion, hitting behavior will be influenced given the difference in the speed between fastballs and curveballs. By manipulating the visual stimulus in this way, our system can present matched (predictive) and mismatched (unpredictive) cues to batters.

Figure 2 shows our experimental setup. When using the system, a batter wears a head mounted display (HTC Vive Pro) and swings with a physical bat (weight: 750 g; length: 84 cm) in response to the pitching stimulus. The frame-rate of the system was 90Hz. The bat has four retro-reflective markers attached to it, and measurements are obtained using 16 motion capture systems (NaturalPoint OptiTrack Prime 13 and Prime 17W) placed around the batting space.

The experiment included ten elite baseball players (10 men, 26.3 $\pm$ 1.4 years, six right-handed batters, and four left-handed batters, a semi-professional baseball team). This study was approved by the ethics committee of our organization, and all participants provided written informed consent. Two types of pitches (fastball at 131.4 km/h, curveball at 100.4 km/h) were presented in the baseball VR experiment. We prepared three different hitting conditions with combinations of the required decision, pitching motion, and pitch type: a condition in which pitching motion and pitch type were

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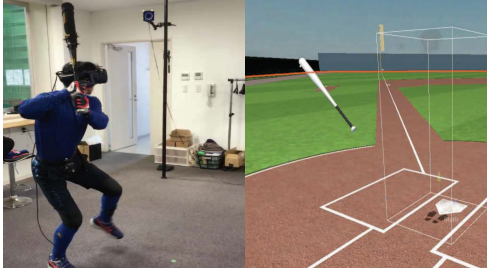


Figure 2: Experimental setup.

matched and the batter was instructed to swing at all pitches (Match-ALL), a condition in which the pitching motion and pitch type were matched and the batter was instructed to swing at fastballs only (Match-Decision), and a condition in which the pitching motion and pitch type were mismatched and the batter was instructed to swing at fastballs only (Mismatch-Decision). Each condition consisted of eight straight pitches and eight curve pitches, for a total of 16 pitches thrown in random order. In the Match-All and Match-Decision conditions, participants practiced four pitches each to familiarize themselves with hitting in a VR environment prior to the main tasks. A total of 56 pitches were presented.

3 RESULT

We compared the Match-All and Match-Decision conditions to assess the effect of decision-making on hitting behavior. Figure 3 shows the mean onset time and mean peak time of the bat swing in fastball trials under both conditions. T-tests revealed no significant difference in the mean onset time ($t(8) = 0.965, p = 0.360$), indicating no significant difference in the decision time for hitting fastballs. In contrast, we observed a significant difference in the mean peak time ($t(8) = 2.681, p = 0.025$), indicating a difference in the timing of ball-hitting. Figure 3 shows the corresponding variability (standard deviation) for onset time and peak time in the fastball trials. There was no significant difference in peak time variability ($t(8) = 2.965, p = 0.015$), while there was a significant difference in the variability of onset time ($t(8) = 1.560, p = 0.153$), indicating that decision timing was stable in the Match-Decision condition.

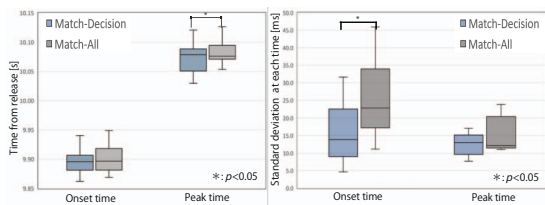


Figure 3: The Match-All/Match-Decision conditions left: Distribution of mean onset time and peak time, right: Distribution of standard deviations for the onset time and peak time

To examine whether batters used predictions based on pitching motion cues to make decisions regarding hitting, we compared the mean onset time and mean peak time of swings against fastballs between the Match-Decision (initial three trials in the session) and Mismatch-Decision (last three trials in the session) conditions, as shown in Figure 4. Both times were significantly delayed in the Mismatch-Decision condition when compared with those in the Match-Decision condition (onset time: $t(8) = 2.433, p = 0.038$, peak time: $t(8) = 2.298, p = 0.047$).

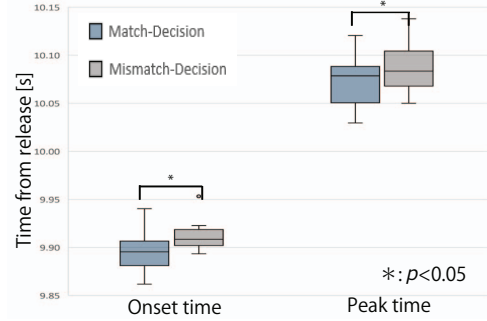


Figure 4: Distribution of mean onset time and mean peak time for the Match-Decision/Mismatch-Decision conditions

4 DISCUSSION

In the Match-All condition, batters were instructed to hit both fastballs and curveballs and did not discriminate between them, whereas discrimination was required to ensure only fastballs were hit in the Match-Decision condition. Therefore, the Match-All condition required batters to prepare and switch between two action plans, while the Match-Decision condition required only a single action plan. Our results reflected this difference in motor planning. We found that the peak time to swing at fastballs occurred later in the Match-All condition than in the Match-Decision condition, which is likely explained by implementation of the plan for slow pitches (curveballs) beforehand. These results emphasize the influence of decision-making strategies on the stability of the decision itself and accuracy of actions.

Our results indicated that, just after the Match-Decision condition, the swing onset time for fastballs was delayed in the Mismatch-Decision condition, in which the pitching motion for curveballs was presented. This is likely due to the batter's prediction of a curveball based on the pitching motion rather than on the trajectory of the ball.

5 CONCLUSION

Using a virtual baseball environment, we manipulated the pitching motion and pitched ball trajectory independently to assess the batters decision-making and motor control in response to predictive and unpredictable cues. Our results indicated that decision-making related to pitch type influenced the stability of the decision itself and accuracy of hitting, and that most batters relied on pitching motion cues rather than on ball trajectory to make decisions. Our VR system may therefore be useful for assessing and training cognitive motor skills in realistic sports scenarios.

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